

## EDWARDS AFB, CA, USA

WMO: 723810

Lat: 34.900N

Lon: 117.867W

Elev: 704

StdP: 93.14

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23114

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -5.7       | -3.7 | -21.1                         | 0.6 | 4.3  | -18.5 | 0.8 | 6.0  | 13.7                  | 10.6 | 12.2 | 10.9 | 1.4                   | 0    | 0.592 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.8                   | 40.6            | 18.1 | 39.1 | 17.6 | 37.8 | 17.2 | 20.5                | 33.8 | 19.7 | 34.2 | 18.9 | 34.0 | 5.4                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.4                            | 12.7 | 24.7 | 14.2 | 11.0 | 25.7 | 12.6 | 9.9 | 25.2 | 62.0          | 33.4 | 58.9 | 34.3 | 56.0 | 33.8 | 24.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 12.6          | 11.5 | 10.5 | DB                         | -9.6  | 43.1               | 2.4 | 1.1                                                | -11.3 | 43.9       | -12.7 | 44.6       | -14.1 | 45.2       | -15.9 | 46.0 | (4) |
| (5)               |      |      | WB                         | -10.9 | 22.2               | 2.4 | 1.0                                                | -12.6 | 22.9       | -14.0 | 23.5       | -15.3 | 24.1       | -17.0 | 24.8 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 17.3  | 7.5   | 9.2   | 12.7  | 15.7  | 19.8  | 25.4  | 28.8  | 28.3  | 24.1  | 17.2  | 11.2  | 6.6  | (6)  |
| (7)  |                                                                                  | DBStd       | 8.45  | 3.18  | 3.41  | 3.50  | 4.15  | 4.15  | 3.94  | 2.49  | 2.40  | 3.47  | 3.58  | 3.67  | 3.48 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 306   | 88    | 51    | 15    | 5     | 0     | 0     | 0     | 0     | 0     | 1     | 30    | 117  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 1548  | 334   | 256   | 176   | 98    | 34    | 3     | 0     | 0     | 4     | 63    | 215   | 364  | (9)  |
| (10) |                                                                                  | CDD10.0     | 2954  | 12    | 29    | 99    | 176   | 304   | 461   | 584   | 568   | 422   | 224   | 66    | 11   | (10) |
| (11) |                                                                                  | CDD18.3     | 1153  | 0     | 0     | 2     | 19    | 79    | 214   | 326   | 309   | 176   | 28    | 1     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 16704 | 1     | 9     | 117   | 412   | 1159  | 3029  | 4589  | 4308  | 2388  | 628   | 63    | 1    | (12) |
| (13) |                                                                                  | CDH26.7     | 9809  | 0     | 1     | 20    | 140   | 546   | 1839  | 2926  | 2744  | 1349  | 238   | 6     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 4.5   | 3.2   | 4.1   | 4.7   | 5.7   | 6.0   | 5.8   | 5.2   | 4.8   | 4.0   | 3.7   | 3.2   | 3.2  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 144   | 28    | 36    | 15    | 8     | 3     | 1     | 4     | 3     | 6     | 7     | 21    |      | (15) |
| (16) |                                                                                  | PrecMax     | 331   | 149   | 152   | 58    | 26    | 24    | 12    | 22    | 11    | 33    | 51    | 27    | 91   | (16) |
| (17) |                                                                                  | PrecMin     | 52    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (17) |
| (18) |                                                                                  | PrecStd     | 86    | 39    | 38    | 16    | 8     | 5     | 3     | 6     | 4     | 9     | 13    | 8     | 23   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 21.3  | 24.6  | 29.0  | 32.7  | 37.1  | 41.7  | 42.7  | 42.1  | 38.6  | 34.0  | 27.5  | 21.5 | (19) |
| (20) |                                                                                  |             | MCWB  | 10.4  | 10.9  | 12.5  | 14.3  | 16.5  | 18.2  | 18.7  | 18.8  | 17.9  | 14.9  | 12.2  | 10.7 | (20) |
| (21) |                                                                                  | 2%          | DB    | 19.0  | 22.4  | 26.4  | 30.3  | 34.3  | 39.3  | 40.8  | 40.1  | 37.1  | 31.7  | 25.2  | 18.3 | (21) |
| (22) |                                                                                  |             | MCWB  | 8.8   | 10.2  | 11.7  | 13.2  | 15.1  | 16.9  | 18.3  | 18.1  | 17.4  | 14.4  | 11.1  | 8.9  | (22) |
| (23) |                                                                                  | 5%          | DB    | 17.2  | 20.0  | 24.2  | 28.0  | 32.3  | 37.5  | 39.2  | 38.8  | 35.7  | 29.5  | 22.8  | 16.2 | (23) |
| (24) |                                                                                  |             | MCWB  | 8.3   | 9.5   | 10.9  | 12.4  | 14.6  | 16.4  | 17.8  | 17.6  | 16.8  | 13.4  | 10.6  | 8.1  | (24) |
| (25) |                                                                                  | 10%         | DB    | 15.1  | 17.4  | 22.0  | 25.6  | 29.9  | 35.7  | 37.7  | 37.4  | 34.0  | 27.3  | 20.1  | 13.9 | (25) |
| (26) |                                                                                  |             | MCWB  | 7.8   | 8.5   | 10.5  | 11.7  | 14.0  | 16.0  | 17.5  | 17.2  | 16.1  | 12.7  | 9.4   | 7.3  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 13.1  | 13.0  | 13.9  | 15.7  | 17.7  | 19.9  | 21.9  | 21.3  | 20.7  | 17.8  | 14.1  | 13.1 | (27) |
| (28) |                                                                                  |             | MCDWB | 16.9  | 19.1  | 23.0  | 28.9  | 34.2  | 37.9  | 32.8  | 34.2  | 32.6  | 27.1  | 20.2  | 16.4 | (28) |
| (29) |                                                                                  | 2%          | WB    | 11.4  | 11.6  | 12.8  | 14.4  | 16.2  | 18.4  | 20.7  | 20.3  | 19.7  | 16.1  | 12.5  | 11.4 | (29) |
| (30) |                                                                                  |             | MCDWB | 15.4  | 18.0  | 22.7  | 26.8  | 31.3  | 35.4  | 33.3  | 34.5  | 30.9  | 26.4  | 21.8  | 15.6 | (30) |
| (31) |                                                                                  | 5%          | WB    | 10.2  | 10.5  | 11.9  | 13.4  | 15.3  | 17.5  | 19.8  | 19.4  | 18.7  | 14.8  | 11.4  | 10.1 | (31) |
| (32) |                                                                                  |             | MCDWB | 14.0  | 17.4  | 21.8  | 25.3  | 29.9  | 34.3  | 34.6  | 35.1  | 30.7  | 26.2  | 20.2  | 14.0 | (32) |
| (33) |                                                                                  | 10%         | WB    | 9.0   | 9.6   | 11.0  | 12.5  | 14.5  | 16.7  | 18.9  | 18.4  | 17.6  | 13.7  | 10.5  | 8.7  | (33) |
| (34) |                                                                                  |             | MCDWB | 13.1  | 15.9  | 20.0  | 23.7  | 28.1  | 32.9  | 34.5  | 34.9  | 30.1  | 24.4  | 18.5  | 12.6 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 14.9  | 14.6  | 15.9  | 15.9  | 16.1  | 17.9  | 17.8  | 18.4  | 18.2  | 18.1  | 17.3  | 14.8 | (35) |
| (36) |                                                                                  |             | MCDBR | 18.8  | 19.6  | 20.4  | 20.2  | 19.5  | 20.0  | 19.3  | 19.6  | 19.8  | 21.6  | 21.1  | 18.0 | (36) |
| (37) |                                                                                  |             | MCWBR | 10.6  | 9.8   | 9.0   | 8.2   | 7.0   | 6.4   | 6.1   | 6.7   | 7.4   | 9.7   | 11.2  | 10.5 | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 12.9  | 15.4  | 17.3  | 17.7  | 18.3  | 18.8  | 16.1  | 17.0  | 16.2  | 17.5  | 17.6  | 13.0 | (38) |
| (39) |                                                                                  |             | MCWBR | 8.0   | 8.1   | 8.0   | 7.3   | 6.7   | 6.4   | 6.1   | 6.5   | 6.1   | 7.8   | 9.5   | 8.2  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.273 | 0.283 | 0.298 | 0.320 | 0.335 | 0.330 | 0.353 | 0.340 | 0.325 | 0.302 | 0.277 | 0.276 | (40) |      |
| (41) |                                                                                  | taud        | 2.548 | 2.538 | 2.485 | 2.424 | 2.405 | 2.432 | 2.414 | 2.429 | 2.464 | 2.515 | 2.560 | 2.556 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 948   | 973   | 979   | 964   | 948   | 947   | 923   | 932   | 935   | 935   | 934   | 923   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 78    | 89    | 102   | 114   | 118   | 115   | 116   | 112   | 103   | 90    | 77    | 74    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 3.11  | 4.05  | 5.52  | 6.83  | 7.71  | 8.39  | 7.89  | 7.40  | 6.31  | 4.77  | 3.54  | 2.83  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.23  | 0.41  | 0.28  | 0.28  | 0.33  | 0.33  | 0.34  | 0.20  | 0.15  | 0.25  | 0.14  | 0.20  | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | -3.04  | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (44 neighbors) |  | +0.61 | N/A     | -1.86  | +0.64   | N/A   | N/A   | N/A         | -112    | +218    | +105    |

Nomenclature: See separate page